Math 4400, Fall 2016 **Quiz 1** Name: ____________________

Calculators, cell phones, or notes are not allowed.

(1) Use Euclid's Algorithm to find the gcd of 168 and 119.

$$\begin{aligned} 168 &= 119 \cdot 1 + 49 \\ 119 &= 49 \cdot 2 + 21 \\ 49 &= 21 \cdot 2 + 7 \\ 21 &= 7 \cdot 3. \end{aligned}$$

So $\gcd(168, 119) = 7$.

(2) Write the continued fraction expansion for $168/119$.

$$1 + \cfrac{1}{2 + \frac{1}{2 + \frac{1}{3}}}.$$

(3) Find all solutions to $168x + 119y = 35$.

$$1 + \frac{1}{2 + \frac{1}{2}} = \frac{7}{5}.$$

Thus $5 \cdot 168 - 7 \cdot 119 = 7$ and hence $25 \cdot 168 - 35 \cdot 119 = 35$. So the solutions are

$$(x, y) = (25, -35) + k(17, -24) \qquad \text{or} \qquad x = 25 + k \cdot 17,\ \ y = -35 - k \cdot 24.$$

The median score was 8.5 out of 10.